NIKHIL GUPTA

DevOps Engineer
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PROFESSIONAL SUMMARY

DevOps engineer with hands-on experience across AWS, Infrastructure as Code, CI/CD and monitoring. Provision AWS infrastructure (EC2, VPC, ALB, S3, CloudWatch) with modular Terraform and remote state; build automated GitHub Actions and Jenkins pipelines that test, scan, containerize with Docker and deploy without manual steps; and run Prometheus and Grafana monitoring. Write Python and Bash for health checks, failover and deployment automation on Linux (Ubuntu) servers running Nginx, with strong Git, security-group and secrets hygiene.

TECHNICAL SKILLS

Cloud (AWS): EC2, VPC, S3, Application Load Balancer, CloudWatch, SNS, Security Groups, Multi-AZ and Multi-Region Architecture
Infrastructure as Code: Terraform (HCL), Remote State Management, Modular Variables/Outputs, Dynamic AMI Lookup
CI/CD & Automation: GitHub Actions, Jenkins (Jenkinsfile Pipelines), Docker Hub, Automated Build-Test-Deploy Workflows
Containers & Orchestration: Docker, Dockerfile Authoring, Image Build & Registry Publishing, Kubernetes (Deployments, Services, Rolling Updates/Rollbacks, Horizontal Pod Autoscaler)
Scripting & Languages: Python, Bash, HCL
Linux & Web Servers: Ubuntu 22.04, Nginx, systemd, Web Server Bootstrapping
Reliability & Practices: Monitoring (Prometheus, Grafana), Health Checks, DR (RTO/RPO) Design, Git/GitHub, pytest, Flask

EXPERIENCE

Python Intern – Editoo Media Services, Gurugram 2023
- Learned Python fundamentals and automation basics, and applied them to real programming tasks under senior guidance.
- Worked on debugging and improving existing scripts, building a practical foundation in reading and fixing other people's code.

MERN Stack Trainee – Grapess Solutions Pvt Ltd, Yamuna Nagar 2024
- Learned MongoDB, Express.js, React.js, and Node.js, and built basic full-stack web applications end-to-end.
- Gained hands-on experience with application architecture and deployment basics, which later carried into infrastructure and DevOps-focused project work.

PROJECTS

Steepline: E-Commerce Site with DevOps Pipeline on AWS | github.com/nikhil-flux/steepline | *Terraform, AWS, Docker, GitHub Actions, Grafana*
- Provisioned 9 AWS resources with Terraform (EC2, Elastic IP, security group, CloudWatch/SNS alerts, budget alarm) hosting a 4-page store at $0 on free tier.
- Built a 3-job GitHub Actions pipeline (lint, build, smoke test, Trivy scan, GHCR push, SSH deploy, health check): zero manual steps per release; fixed 2 critical CVEs it caught.
- Monitored 5 containerized services (CPU, memory, requests) with Prometheus, Grafana and exporters.

Python CI/CD Pipeline with Docker & GitHub Actions | github.com/nikhil-flux/python-cicd-demo | *Python, Flask, pytest, Docker, GitHub Actions, Docker Hub*
- Built a Flask REST API with a health-check endpoint, backed by an automated pytest suite that runs on every push.
- Automated the full build-test-deploy cycle with a GitHub Actions workflow that containerizes the app with Docker and publishes the image to Docker Hub.

Multi-Region Disaster Recovery on AWS | github.com/nikhil-flux/aws-multi-region-disaster-recovery | *Terraform, AWS (EC2, ALB, VPC, S3), Python*
- Provisioned matching multi-AZ infrastructure across 2 AWS regions (us-east-1, us-west-2) with Terraform, covering VPCs, EC2 (Nginx), and Application Load Balancers.
- Designed a cold-standby DR strategy targeting a 5-10 minute RTO, with versioned S3 buckets replicated across both regions.
- Automated failover detection with a Python health-check script that continuously monitors the primary region's load balancer.

AWS Infrastructure as Code with Terraform | github.com/nikhil-flux/aws-terraform-infrastructure | *Terraform, AWS VPC, EC2, S3*
- Built a modular Terraform codebase provisioning VPC, EC2, and S3 resources, organized into reusable files with variables and outputs.
- Configured a remote S3 state backend for safer, team-shareable Terraform state management.

Automated Nginx Web Server on EC2 | github.com/nikhil-flux/terraform-aws-nginx-ec2 | *Terraform, AWS EC2, Bash, Nginx*
- Automated end-to-end deployment of an Ubuntu 22.04 EC2 instance with a locked-down security group (HTTP/HTTPS/SSH) and Nginx installed via a first-boot Bash user-data script.
- Parameterized region, instance type, VPC, subnet, and SSH CIDR as Terraform variables, with dynamic AMI lookup and secrets kept out of version control.

Jenkins CI/CD Pipeline | github.com/nikhil-flux/jenkins-demo | *Jenkins, Docker, Git, GitHub*
- Deployed Jenkins in a Docker container and connected it to GitHub to trigger a 4-stage Jenkinsfile pipeline (checkout, build, test, deploy) automatically on every push.

EDUCATION

B.Tech, Information Technology – Ganpati Institute of Technology and Management 2020 – 2024
Class XII – Guru Nanak Shiksha Niketan, Kurukshetra 2019 – 2020
Class X – Manu Sikhsha Sadan, Kurukshetra 2018 – 2019